Quiz # 5 (Data Structures) CLO2 **BS DS Fall 2025 Morning**

Name:_____ **Roll number:_____**

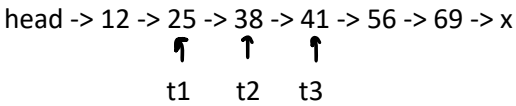
Time Allowed: 15 min **Total Marks: 30**

Q1: For a singly linked list, swap the data values of alternate adjacent node pairs (first with second, third with fourth, and so on).

```
template <class T>
void LinkedList<T>::swapAlternatePairs()
{
   Node<T>* temp = head;

   while (temp != NULL && temp->next != NULL)
   {
      T value = temp->info;
      temp->info = temp->next->info;
      temp->next->info = value;
      temp = temp->next->next;
   }
}
```

Q2. Consider the following linked list with three pointers, t1, t2, and t3, pointing to particular nodes. Perform the operations below, redraw the list, and write the output of the print statement. Ignore any logical issues.




```
t1->next = t3;
t2->next = head;
head->next = t2;
t3->next = t1;

for (t1 = head; t1 != NULL; t1 = t1->next)
   cout << t1->data << ' ';
```

Output:

__12 38 12 38 12 38 . . . _____

New List:

_head → 12 → 38 → 12 → 38 → 12 → ....

Q3: Consider the following two function part of Linear singly link list. What is the output of the given main function.

```
template <class T>
void LSLL<T>::F4()
{
   head = F4Helper(head);
}
template <class T>
Node<T>* LSLL<T>::F4Helper(Node<T>* node)
{
   if (node == NULL)
      return NULL;

   if (node->data % 2 == 0)
   {
      Node<T>* temp = node->next;
      delete node;
      return F4Helper(temp);
   }
   node->next = F4Helper(node->next);
   return node;
}
```

```
int main()
{

   LSLL<int> list;
   list.insertAtHead(23);
   list.insertAtTail(8);
   list.insertAtHead(3);
   list.insertAtTail(4);
   list.insertAtHead(6);
   list.insertAtTail(7);
   list.insertAtHead(9);

   list.F4 ();
   list.display();
return 0;

}
```

9 3 23 7